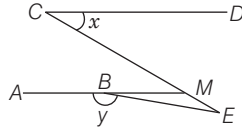


35. (d) We produced line AB to meet line CE at M .



Since, $AM \parallel CD$.

$$\angle DCM = \angle BMC = x$$

[alternate angles]

CME is a straight line

$$\therefore \angle BME = \pi - x$$

Also, ABM is a straight line.

$$\therefore \angle EBM = \pi - y$$

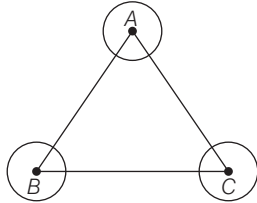
Now, in $\triangle BEM$

$$\angle EBM + \angle BME + \angle BEM = \pi$$

$$\Rightarrow \pi - y + \pi - x + \angle E = \pi$$

$$\Rightarrow \angle E = x + y - \pi$$

36. (c) $\therefore \angle A = 360^\circ - \text{Ext } \angle A$

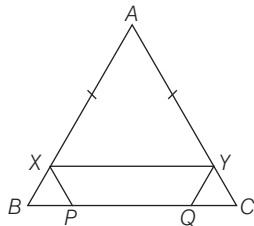


Similarly, $\angle B = 360^\circ - \text{Ext } \angle B$ and $\angle C = 360^\circ - \text{Ext } \angle C$

We know that, the sum of all the angles of a triangle is 180° .

$$\begin{aligned} \therefore \angle A + \angle B + \angle C &= 180^\circ \\ \Rightarrow 360^\circ - \text{Ext } \angle A + 360^\circ - \text{Ext } \angle B &+ 360^\circ - \text{Ext } \angle C = 180^\circ \\ \Rightarrow \text{Ext } \angle A + \text{Ext } \angle B + \text{Ext } \angle C &= 1080^\circ - 180^\circ = 900^\circ \end{aligned}$$

37. (d) Since, $XP \parallel AC$ and $YQ \parallel AB$



$$\therefore \angle XBP = \angle YQC \text{ and } \angle XPB = \angle YCQ$$

Also, $XY \parallel BC$

$\therefore XYQB$ and $XYCP$ are parallelogram

$$\Rightarrow XB = YQ \text{ and } XP = YC$$

So, $\triangle XBP$ and $\triangle YCQ$ are congruent triangles.

Now, $XY \parallel BC$

$$\therefore \frac{AX}{AB} = \frac{XY}{BC} \Rightarrow AX = XY$$

$$[\because AB = BC = 30 \text{ cm}]$$

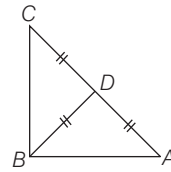
$$\begin{aligned} \text{Also, } XY + XP + YQ &= 40 \text{ [given]} \\ \Rightarrow AX + XB + YQ &= 40 \\ &[\because XY = AX, XP = XB] \\ \Rightarrow AB + YQ &= 40 \\ \Rightarrow YQ &= 40 - 30 = 10 \text{ cm} \\ \therefore YQ &= XP = 10 \text{ cm} \\ \therefore BP &= CQ = 10 \text{ cm} \\ \therefore PQ &= 30 - BP - CQ \\ &= 30 - 10 - 10 = 10 \text{ cm} \end{aligned}$$

38. (b) As, $OP \parallel YZ$
 $\Rightarrow \angle POY = \angle OYZ$ [alternate angles]
 $\Rightarrow \angle PYO = \angle POY$
 [since OY is angle bisector of $\angle Y$]
 $\therefore PY = PO$
 As $\angle XYZ$ is an acute angle.
 $\therefore \frac{1}{2} \angle XYZ < 45^\circ$
 $\therefore \angle POY = \angle PYO < 45^\circ$
 $\therefore \angle YPO > 90^\circ$
 Hence, $\triangle PYO$ is an isosceles triangle but not a right angled triangle.

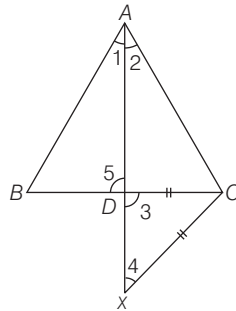
39. (c) Here, we see that

$$CD = BD = DA$$

This is possible only when ABC is right angled triangle.



40. (a) In $\triangle DCX$, $CD = CX$ [given]
 $\angle 3 = \angle 4$
 [opposite angle of same sides]
 But $\angle 3 = \angle 5$, So, $\angle 4 = \angle 5$
 In $\triangle ABD$ and $\triangle ACX$,
 $\angle 1 = \angle 2$ [given]



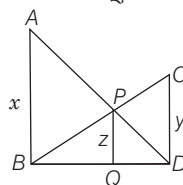
$$\begin{aligned} \angle 4 &= \angle 5 \\ \therefore \triangle ABD &\sim \triangle ACX \end{aligned}$$

[by AA similarity]

41. (a) Since, $AB \perp BD$ and $PQ \perp BD$

$$\Rightarrow AB \parallel PQ$$

$$\text{So, } \triangle ABD \sim \triangle PQD$$



$$\text{Then, } \frac{x}{z} = \frac{BD}{QD} \quad \dots(i)$$

Since $CD \parallel PQ$

So, $\triangle BCD \sim \triangle BPQ$,

$$\therefore \frac{z}{y} = \frac{BQ}{BD}$$

$$\Rightarrow \frac{z}{y} = \frac{BD - QD}{BD}$$

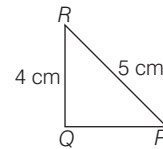
$$\Rightarrow \frac{z}{y} = 1 - \frac{QD}{BD}$$

$$\Rightarrow \frac{z}{y} = 1 - \frac{z}{x}$$

$$\Rightarrow \frac{z}{y} = 1 - \frac{z}{x} \quad [\text{from Eq. (i)}]$$

$$\Rightarrow \frac{z}{x} + \frac{z}{y} = 1 \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

42. (d) In right angle $\triangle PQR$, by pythagoras theorem,



$$QP^2 = (5)^2 - (4)^2 = 9 \Rightarrow QP = 3 \text{ cm}$$

In second $\triangle ABC$ whose sides are 3 cm, 4 cm and 5 cm. So, the sides of both triangle are same, hence they are congruent.

43. (b) Given, $AC = BC$

$$\therefore \angle ABC = \angle BAC$$

[angles of an isosceles triangle]

$$\Rightarrow \angle BAC = \angle ABC = 38^\circ$$

In $\triangle ABC$,

$$\angle ACB = 180^\circ - (\angle BAC + \angle ABC)$$

[by angle sum property of a triangle]

$$= 180^\circ - (38^\circ + 38^\circ)$$

$$= 180^\circ - 76^\circ = 104^\circ$$

$$\text{In } \triangle ACD, \angle ACD = 180^\circ - 104^\circ = 76^\circ$$

[by linear pair]

$$\text{and } \angle CAD = \angle ACD = 76^\circ$$

[$\because CD = AD$]

$$\therefore \angle ADC = 180^\circ - (\angle CAD + \angle ACD)$$

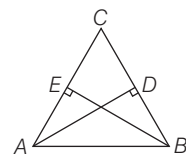
$$= 180^\circ - (76^\circ + 76^\circ) = 28^\circ$$

44. (c) I. It is true that the three medians of a triangle divide it into six triangles of equal area.

II. It is also true that, the perimeter of a triangle is greater than the sum of the lengths of its three medians.

Hence, I and II are correct.

45. (d) In $\triangle CAD$ and $\triangle CBE$,



$\angle C = \angle C$ [common]
 $\angle CEB = \angle ADC$ [each 90°]
 $\therefore \triangle CAD \sim \triangle CBE$ [by AA similarity]

Sides will be in same proportion.

$$\frac{CA}{CB} = \frac{CD}{CE} \text{ and } \frac{AD}{BE} = \frac{CD}{CE}$$

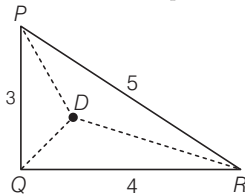
Hence, all three statements are correct.

46. (b) Given, $PQ = 3$ cm, $QR = 4$ cm and $RP = 5$ cm

Here, $RP^2 = PQ^2 + QR^2$

So, PQR is a right angled triangle.

Let D is an interior point in $\triangle PQR$



- In $\triangle DPQ$, $DP + DQ > 3$... (i)
 similarly, $DQ + DR > 4$... (ii)
 and $DP + DR > 5$... (iii)

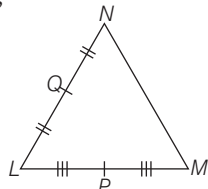
adding (i), (ii) and (iii), we get

$$2(DP + DQ + DR) > 12$$

$$\Rightarrow DP + DQ + DR > 6$$

Hence, both statements I and II are individually true and statement II is not correct explanation of statement I.

47. (b) Given,



- I. $PQ^2 = MP^2 + NQ^2$
 $\Rightarrow PQ^2 = LP^2 + LQ^2$
 [$\because LP = MP$ and $NQ = LQ$]
 $\Rightarrow \angle QLP = 90^\circ$
 It means, $\triangle NLM$ is a right angled triangle.

- II. It also true that if in a $\triangle ABC$, $AB^2 > BC^2 + CA^2$, then $\angle ACB$ is obtuse.

Hence, both statements are individually true but statement II is not the correct explanation of statement I.

48. (c) Given that,

$$\frac{1}{2}\angle A + \frac{1}{3}\angle C + \frac{1}{2}\angle B = 80^\circ$$

$$\Rightarrow 3\angle A + 2\angle C + 3\angle B = 480^\circ$$

$$\Rightarrow 3(\angle A + \angle B) + 2\angle C = 480^\circ \dots (i)$$

Also, in $\triangle ABC$,

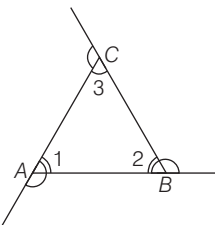
$$\angle A + \angle B + \angle C = 180^\circ$$

$$\Rightarrow 3(\angle A + \angle B) + 2\angle C = 540^\circ \dots (ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$\angle C = 60^\circ$$

49. (c)



By Exterior angle theorem,

$$\text{ext. } \angle A = \angle 2 + \angle 3$$

$$\text{ext. } \angle B = \angle 1 + \angle 3$$

$$\text{ext. } \angle C = \angle 1 + \angle 2$$

Sum of the three exterior angles

$$= (\angle 1 + \angle 2) + (\angle 2 + \angle 3) + (\angle 3 + \angle 1)$$

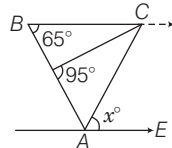
$$= 2(\angle 1 + \angle 2 + \angle 3)$$

$$= 2 \times 180^\circ = 360^\circ$$

$$[\because \angle 1 + \angle 2 + \angle 3 = 180^\circ]$$

50. (d) Given that, $BC \parallel AE$

$$\angle CBA + \angle EAB = 180^\circ$$



$$\Rightarrow \angle EAB = 180^\circ - 65^\circ = 115^\circ$$

$$\therefore BC = AC$$

Hence, $\triangle ABC$ is an isosceles triangle.

$$\Rightarrow \angle CBA = \angle CAB = 65^\circ$$

$$\text{Now, } \angle EAB = \angle EAC + \angle CAB$$

$$\Rightarrow 115^\circ = x + 65^\circ \Rightarrow x = 50^\circ$$

51. (b) Given that, $\angle ABC = 65^\circ$

$$\text{and } \angle CDE = 15^\circ$$

Since, $AB \parallel CD$

$$\angle DCB = \angle ABC = 65^\circ$$

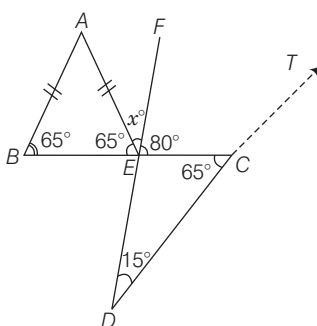
[alternate angles]

$$\text{Also, } \angle DCE = \angle DCB \text{ [same angle]}$$

$$\text{Now, } \angle FEC = \angle CDE + \angle DCE$$

[exterior angle]

$$15^\circ + 65^\circ = 80^\circ$$



$$\therefore \angle DEC + \angle FEC = 180^\circ \text{ [linear pair]}$$

Given that, $AB = AE$

$$\therefore \angle ABE = \angle AEB = 65^\circ$$

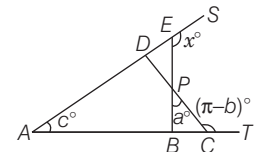
$$\therefore \angle AEB + \angle AEF + \angle FEC = 180^\circ$$

[straight line]

$$\Rightarrow 65^\circ + x^\circ + 80^\circ = 180^\circ$$

$$\therefore x^\circ = 180^\circ - 145^\circ = 35^\circ$$

52. (c) $\therefore \angle PCT + \angle PCB = \pi$ [linear pair]



$$\angle PCB = \pi - (\pi - b^\circ) = b^\circ \dots (i)$$

In $\triangle BPC$, $\angle PCB + \angle BPC + \angle PBC = \pi$

[by angle sum property of a triangle]

$$\Rightarrow \angle PBC = \pi - \angle PCB - \angle BPC$$

$$= \pi - b^\circ - a^\circ \dots (ii)$$

$$\therefore \angle ABE + \angle EBC = \pi \text{ [linear pair]}$$

$$\Rightarrow \angle ABE = \pi - \angle PBC$$

$$[\because \angle PBC = \angle EBC]$$

$$= \pi - (\pi - b^\circ - a^\circ) = a^\circ + b^\circ \dots (iii)$$

Now, in $\triangle ABE$

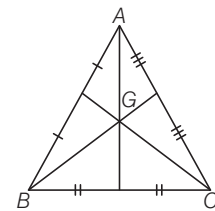
Sum of two interior angles = Exterior angle

$$\angle EAB + \angle ABE = \angle BES$$

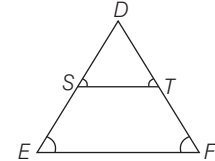
$$c^\circ + b^\circ + a^\circ = x^\circ$$

$$\therefore x^\circ = a^\circ + b^\circ + c^\circ$$

53. (d) $GA = GB = GC$ is true only and only for equilateral triangle and here it is not given that ABC is an equilateral triangle. So, statement I is not correct. Similarly, statement II will also hold only for equilateral triangle.



54. (a) Given, $\frac{DS}{DE} = \frac{1}{\sqrt{2}}$



$$\therefore DE = \sqrt{2} \times DS$$

Since, ST is parallel to EF .

$$\therefore \angle S = \angle E \text{ and } \angle T = \angle F$$

So, $\triangle DST$ and $\triangle DEF$ are similar.

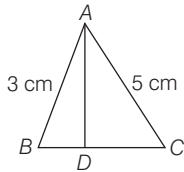
Ratio of areas = Ratio of squares of corresponding sides.

$$\Rightarrow \frac{\Delta DST}{\Delta DEF} = \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{2}$$

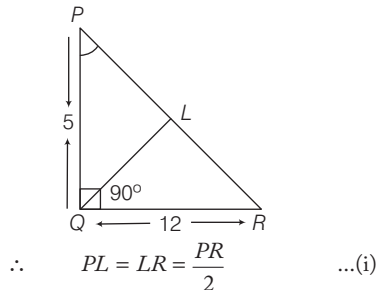
$$\Rightarrow \Delta DEF = 2 \Delta DST$$

Hence, both statements I and II are true and statement II is the correct explanation of statement I.

55. (c) We know that, The sum of any two sides of a triangle is greater than twice the median drawn to the third side.
 i.e. $(AB + AC) > 2AD$
 $\Rightarrow (3 + 5) > 2AD \Rightarrow AD < 4$
 Hence, AD is always less than 4 cm.



56. (d) Given that, $PQ = 5$ cm, $QR = 12$ cm and QL is a median.



$$\begin{aligned} \therefore PL = LR = \frac{PR}{2} \quad \dots(i) \\ \text{In } \Delta PQR, (PR)^2 = (PQ)^2 + (QR)^2 \\ \text{[by pythagoras theorem]} \\ (PR)^2 = (5)^2 + (12)^2 \\ (PR)^2 = 25 + 144 \Rightarrow 169 = (13)^2 \\ \Rightarrow PR^2 = (13)^2 \Rightarrow PR = 13 \end{aligned}$$

Now, by mid-point theorem, if L is the mid-point of the hypotenuse PR of a right angled ΔPQR , then

$$QL = \frac{1}{2} PR = \frac{1}{2} (13) = 6.5 \text{ cm}$$

57. (d) In ΔABC and ΔACD ,
 $\therefore \frac{AC}{AB} = \frac{4}{AC}$
 $\therefore AC^2 = 4 \times 13 = 52$... (i)
 $[\because AB = 4 + 9 = 13]$

$$\text{In } \Delta ABC \text{ and } \Delta BCD, \frac{BC}{AB} = \frac{9}{BC}$$

$$\begin{aligned} \Rightarrow BC^2 = 9 \times 13 = 117 \quad \dots(ii) \\ \text{Now, } \frac{1}{CD^2} = \frac{1}{AC^2} + \frac{1}{BC^2} = \frac{1}{52} + \frac{1}{117} \end{aligned}$$

$$= \frac{9 + 4}{13 \times 4 \times 9} \Rightarrow \frac{1}{CD^2} = \frac{1}{36}$$

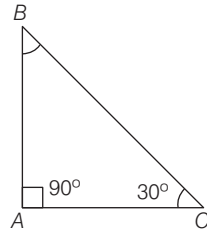
$$\therefore CD = 6 \text{ cm}$$

Shortcut Method

$$CD^2 = AD \cdot DB$$

$$CD = \sqrt{AD \cdot DB} = \sqrt{4 \times 9} = \sqrt{36} = 6 \text{ cm}$$

58. (d) Given that, $\angle A = 90^\circ$ and $\angle C = 30^\circ$



$\therefore ABC$ is a right angled triangle.

Also, given that $BC = 2AB$

$$\Rightarrow AB = \frac{BC}{2} \quad \dots(i)$$

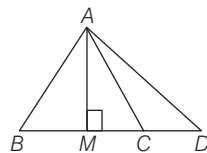
By pythagoras theorem,

$$\begin{aligned} BC^2 &= AC^2 + AB^2 \\ \Rightarrow (2AB)^2 &= AC^2 + AB^2 \\ \Rightarrow AC^2 &= 4AB^2 - AB^2 = 3AB^2 \\ \Rightarrow AC &= \sqrt{3} \cdot AB = \frac{\sqrt{3}}{2} \cdot (2AB) \\ \therefore AC &= \frac{\sqrt{3}}{2} \cdot BC \quad \text{[from Eq.(i)]} \end{aligned}$$

59. (d) Draw $AM \perp BC$

Let $AB = BC = AC = x$

Then, $BM = MC = \frac{x}{2}$



$$\text{Also, } CD = \frac{x}{2} \quad [\because BC = 2CD]$$

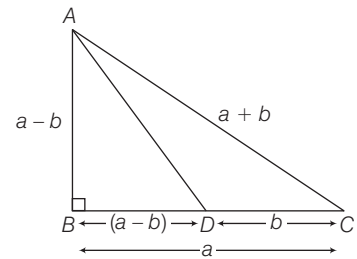
Now, in ΔAMC , by pythagoras theorem,

$$\begin{aligned} AM^2 &= AC^2 - MC^2 \\ &= x^2 - \frac{x^2}{4} = \frac{3x^2}{4} \end{aligned}$$

In ΔAMD , by pythagoras theorem,

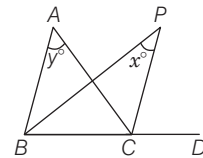
$$\begin{aligned} AD^2 &= AM^2 + MD^2 \\ &= \frac{3x^2}{4} + x^2 = \frac{7x^2}{4} \\ &= 7 \left(\frac{x^2}{4} \right) = 7CD^2 \quad \left[\because CD = \frac{x}{2} \right] \end{aligned}$$

60. (d) In right angle ΔABC ,
 By pythagoras theorem,
 $(a + b)^2 = (a - b)^2 + a^2$



$$\begin{aligned} \Rightarrow a^2 + b^2 + 2ab &= a^2 + b^2 - 2ab + a^2 \\ \Rightarrow 4ab &= a^2 \Rightarrow 4b = a \\ \text{Now, } \frac{BD}{DC} &= \frac{a-b}{b} \\ &= \frac{4b-b}{b} = \frac{3b}{b} = \frac{3}{1} \\ \text{or } 3 : 1 \end{aligned}$$

61. (d) Given that, $\angle BPC = x^\circ$
 and $\angle BAC = y^\circ$



Since, BP and CP are the angle bisectors of $\angle ABC$ and $\angle ACD$, respectively.

$$\therefore \angle ABC = 2\angle PBC \quad \dots(i)$$

$$\text{and } \angle ACD = 2\angle PCD \quad \dots(ii)$$

since sum of two interior angles is equal to exterior angle

$$\therefore \angle PCD = x^\circ + \angle PBC \quad \dots(iii)$$

$$\text{and } \angle ACD = y^\circ + \angle ABC$$

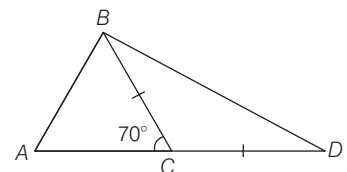
$$\Rightarrow 2\angle PCD = y^\circ + 2\angle PBC$$

[using (i) and (ii)]

$$\Rightarrow 2[x^\circ + \angle PBC] = y^\circ + 2\angle PBC$$

$$\Rightarrow 2x^\circ = y^\circ$$

62. (a) $\therefore \angle ACB + \angle BCD = 180^\circ$ [linear pair]
 $\angle BCD = 180^\circ - 70^\circ = 110^\circ$



In ΔBCD , $BC = CD$

$$\Rightarrow \angle CBD = \angle CDB \quad \dots(i)$$

[angles opposite to equal side]

Also, $\angle BCD + \angle CBD + \angle CDB = 180^\circ$

[by angle sum property of a triangle]

$$2\angle CDB = 180^\circ - \angle BCD$$

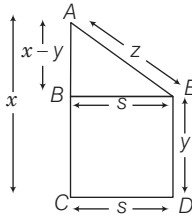
$$= 180^\circ - 110^\circ = 70^\circ$$

$$\therefore \angle CDB = \angle ADB = \frac{70^\circ}{2} = 35^\circ$$

63. (d) Now, applying pythagoras theorem in $\triangle ABE$,

$$AE^2 = AB^2 + BE^2$$

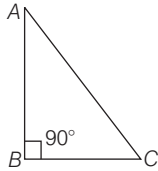
$$\Rightarrow z^2 = (x-y)^2 + s^2$$



$$\Rightarrow z^2 = x^2 + y^2 - 2xy + s^2$$

$$\therefore s^2 = z^2 - x^2 - y^2 + 2xy$$

64. (b) Given, $\angle C = 2\angle A$ and $\angle B = 90^\circ$
 $\therefore \angle A + \angle C = 90^\circ$



$$\Rightarrow \angle A + 2\angle A = 90^\circ \quad [\because \angle C = 2\angle A]$$

$$\Rightarrow 3\angle A = 90^\circ \Rightarrow \angle A = 30^\circ$$

So, $\angle C = 90^\circ - 30^\circ = 60^\circ$

Now, if the angles of a triangle are of measure 30° , 60° and 90° .

Then, side opposite to 30° i.e.

$$BC = \frac{1}{2} \times \text{hypotenuse}$$

$$\Rightarrow BC = \frac{1}{2} AC \Rightarrow AC = 2BC$$

In right angled $\triangle ABC$,

by pythagoras theorem

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow 4BC^2 = AB^2 + BC^2 \quad [\because AC = 2BC]$$

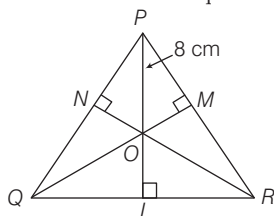
$$\Rightarrow AB^2 = 4BC^2 - BC^2 = 3BC^2$$

65. (d) Since, PQR is an equilateral triangle. Then, PL is also the median of $\triangle PQR$. Similarly, RN and QM are also the median of $\triangle PQR$ and O is the centroid.

$$\text{So, } \frac{PO}{OL} = \frac{2}{1} \Rightarrow OL = \frac{PO}{2} = \frac{8}{2} = 4 \text{ cm}$$

$$\text{Now, altitude of } \triangle PQR = \frac{\sqrt{3}a}{2}$$

[where, a = length of side of an equilateral $\triangle PQR$]



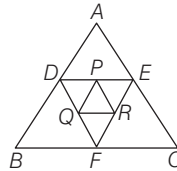
$$PO + OL = \frac{\sqrt{3}a}{2}, \quad 8 + 4 = \frac{\sqrt{3}a}{2}$$

$$a = \frac{12 \times 2}{\sqrt{3}} = \frac{24}{\sqrt{3}} \text{ cm}$$

$$\therefore \text{Perimeter of } \triangle PQR = 3a$$

$$= \frac{3 \times 24}{\sqrt{3}} = 24\sqrt{3} \text{ cm}$$

66. (b) Perimeter of $\triangle PQR = 1+2+3=6$ units



$\therefore P, Q, R$ are the mid-points of DE, DF and FE of $\triangle DEF$.

$$\therefore 2PQ = FE$$

[by mid-point theorem]

Similarly, $DF = 2PR$ and $DE = 2QR$

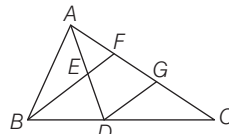
$$\therefore \text{Perimeter of } \triangle DEF = 2 \times 6 = 12 \text{ units}$$

$$\text{Similarly, perimeter of } \triangle ABC$$

$$= 2 \times \text{perimeter of } \triangle DEF$$

$$= 2 \times 12 = 24 \text{ units}$$

67. (b) Draw a line segment DG parallel to BF . Then, in $\triangle ADG$,



$$BF \parallel DG$$

$$\therefore EF \parallel DG$$

$$\text{and } AE = ED$$

[since, E is mid-point of AD]

$$\therefore AF = FG \quad \dots(i)$$

Similarly, in $\triangle BCF$

$$DG \parallel BF \text{ and } BD = DC$$

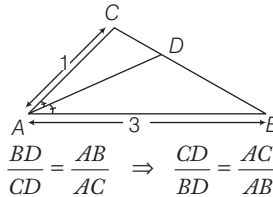
$$\therefore FG = GC \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$CF = \frac{2}{3} AC$$

68. (d) In $\triangle ABC$, AD is the internal angle bisector of $\angle A$

Using property of internal angle bisector.



On adding both sides, we get

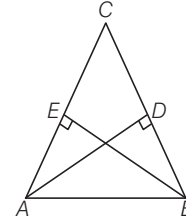
$$\Rightarrow \frac{CD}{BD} + 1 = \frac{AC}{AB} + 1$$

$$\Rightarrow \frac{CD + BD}{BD} = \frac{AC + AB}{AB}$$

$$\Rightarrow \frac{BC}{BD} = \frac{3+1}{3} \Rightarrow \frac{BD}{BC} = \frac{3}{4}$$

$$\therefore BD : BC = 3 : 4$$

69. (b) In $\triangle ADC$ and $\triangle BEC$



$$\angle BEC = \angle ADC = 90^\circ$$

$$\angle ACD = \angle BCE \quad [\text{common}]$$

$\therefore \triangle ADC \sim \triangle BEC$ [by AA similarity]

$$\frac{AC}{BC} = \frac{CD}{CE}$$

$$\Rightarrow AC \times CE = BC \times CD$$

$$\Rightarrow CE \times CA = CD \times CB$$

70. (b) Let the sides of triangle be $\frac{a}{r}, a, ar$ and since $r < 1$.

$$\therefore \frac{a}{r} > a > ar$$

Now, triangle is right angled.

Using pythagoras theorem,

$$\left(\frac{a}{r}\right)^2 = (a)^2 + (ar)^2$$

$$\Rightarrow \frac{a^2}{r^2} = a^2 + a^2 r^2$$

$$\Rightarrow \frac{a^2}{r^2} = a^2(1 + r^2) \Rightarrow r^2 + r^4 = 1$$

$$\text{Put } r^2 = x,$$

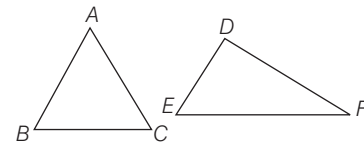
$$\Rightarrow x^2 + x - 1 = 0$$

$$x = \frac{-1 \pm \sqrt{1 - 4(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

Since, sides of triangle cannot be negative

$$\therefore r^2 = \frac{\sqrt{5} - 1}{2}$$

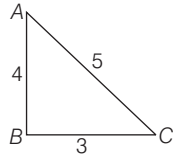
71. (a) $\triangle ABC \sim \triangle DEF$



$$\therefore \frac{AB}{DE} = \frac{BC}{EF} \Rightarrow \frac{1}{2} = \frac{8}{EF}$$

$$EF = 16 \text{ cm}$$

72. (a) Since, the triangle is right angled. So, all the three consecutive sides must satisfy pythagoras theorem.

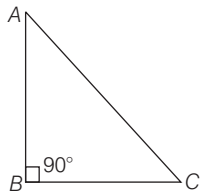


Hence, 3, 4 and 5 are the sides of triangle which satisfy pythagoras theorem.

$$\therefore \text{Area of triangle} = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$$

73. (a) In a triangle, Sum of two sides is always greater than the third side
i.e. $x < 40$... (i)
Difference of two sides is always less than third side
i.e. $10 < x$... (ii)
From Eqs. (i) and (ii), $10 < x < 40$

74. (a) Given, in $\triangle ABC$, $\angle B = 2\angle C = 2\angle A$
We know that, sum of all angles of a triangle = 180°
 $\Rightarrow \angle A + \angle B + \angle C = 180^\circ$
 $\Rightarrow \angle A + 2\angle A + \angle A = 180^\circ$
 $[\because 2\angle C = 2\angle A = \angle B]$
 $\Rightarrow 4\angle A = 180^\circ$
 $\Rightarrow \angle A = \frac{180^\circ}{4} = 45^\circ$



$$\therefore \angle B = 90^\circ \text{ and } \angle C = 45^\circ$$

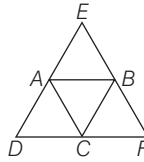
Thus, $\triangle ABC$ is a right angled triangle, right angle at B and $AB = BC$.

In $\triangle ABC$, by pythagoras theorem,

$$\begin{aligned} AB^2 + BC^2 &= AC^2 \\ \Rightarrow AB^2 + AB^2 &= AC^2 & [\because AB = BC] \\ \Rightarrow 2AB^2 &= AC^2 \\ \Rightarrow \sqrt{2}AB &= AC \\ &[\text{taking square root on both sides}] \\ \Rightarrow \frac{AC}{AB} &= \frac{\sqrt{2}}{1} \\ \therefore AC : AB &= \sqrt{2} : 1 \end{aligned}$$

75. (c) I. On drawing the three straight lines through the three vertices of $\triangle ABC$, we get the following figure
Here, $AB \parallel DF$, $BC \parallel DE$ and $AC \parallel EF$

Obviously, A, B and C are the mid-points of DE, EF and DF, respectively.



By mid-point theorem,

$$BC = \frac{1}{2} DE \text{ or } DE = 2BC$$

Similarly, $DF = 2AB$
and $EF = 2AC$

Hence, statement I is correct.

II. Also, area of

$$\triangle ABC = \frac{1}{4} \text{ area of } \triangle DEF$$

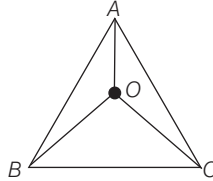
or area of $\triangle DEF = 4$ area of $\triangle ABC$

Hence, statement II is also correct.

76. (c) See solved answer of 67.

77. (c) Given, O is equidistant from AB, BC and AC

$\therefore$ O is the incentre of $\triangle ABC$



Join OA, OB and OC.

In $\triangle ABC$, OA, OB and OC are angle bisectors.

$$\therefore \angle OAB = \angle OAC = \frac{1}{2} \angle BAC$$

$$\therefore \angle OBA = \angle OBC = \frac{1}{2} \angle ABC$$

$$\text{and } \angle OCB = \angle OCA = \frac{1}{2} \angle ACB$$

Now, in $\triangle ABC$,

$$\angle ABC + \angle BAC + \angle ACB = 180^\circ$$

$[\because \text{angle sum property}]$

$$\therefore \angle OAC + \angle OCB + \angle OBA = 90^\circ$$

[option]

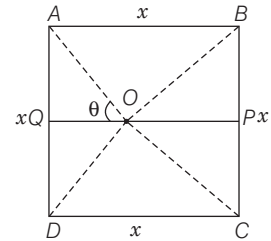
II. This statement is false as

$$\angle BOC = 90^\circ + \frac{1}{2} \angle BAC$$

III. The statement perpendiculars drawn from any point on OA to AB and AC are always equal so it is true, because O is equidistant from AB and AC.

Hence, the statements I and III are correct.

78. (d)



Let side of square ABCD be x and drawn equilateral $\triangle BOC$ inside ABCD.

Such that, $BO = OC = BC = x$

$$\text{and } BP = \frac{BC}{2} = \frac{x}{2}$$

$$\begin{aligned} \therefore PO &= \sqrt{x^2 - \left(\frac{x}{2}\right)^2} = \sqrt{x^2 - \frac{x^2}{4}} \\ &= \sqrt{\frac{3x^2}{4}} = \frac{\sqrt{3}}{2}x \end{aligned}$$

$$\text{and } OQ = AB - PO = x - \frac{\sqrt{3}}{2}x$$

$$= x \left(\frac{2 - \sqrt{3}}{2} \right)$$

Since, $\angle AOD = 90^\circ$

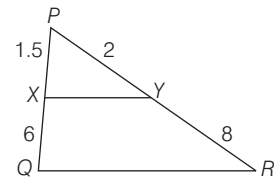
$$\Rightarrow \angle AOQ = \theta$$

In $\triangle AOQ$,

$$\begin{aligned} \tan \theta &= \frac{AQ}{OQ} = \frac{AD}{OQ} = \frac{\frac{x}{2}}{x \left(\frac{2 - \sqrt{3}}{2} \right)} \\ &= \frac{x}{2} \times \frac{2}{x(2 - \sqrt{3})} \\ &= \frac{1}{2 - \sqrt{3}} \times \frac{2 + \sqrt{3}}{2 + \sqrt{3}} = \frac{2 + \sqrt{3}}{4 - 3} \\ &[\because (a + b)(a - b) = a^2 - b^2] \\ &= 2 + \sqrt{3} \end{aligned}$$

79. (d) In $\triangle PQR$,

$$\frac{PX}{XQ} = \frac{1.5}{6} = \frac{1}{4} \text{ and } \frac{PY}{YR} = \frac{2}{8} = \frac{1}{4}$$



$$\text{So, } \frac{PX}{XQ} = \frac{PY}{YR}$$

$$\text{I. } \frac{PX}{XY} = \frac{PQ}{QR} \Rightarrow \frac{1.5}{XY} = \frac{7.5}{QR}$$

$$\Rightarrow QR = 5XY$$

$$\text{II. Also, } \frac{PX}{PQ} = \frac{PY}{PR} = \frac{1}{5}$$

$$\Rightarrow QR \text{ is parallel of } XY.$$